Task 1: Conceptual Understanding (Mandatory - Written)

Q.1 What is the difference between "Love" and "love" in NLP?

= In Natural Language Processing, "Love" and "love" are treated as different tokens if the ~~text~~ text is case sensitive.

"Love" (uppercase) may represent:
- A proper noun
- The begining of sentence

If preprocessing does not normalise case, the model may treat them as two separate words,

Q.2 What happens if stopwords are not removed?

= stopwords are common words like is, the, and, in, on.
- Noise increases.
- Model performance decreases.
- Higher computations costs.
- Diluted features.

Q.3 Provide two real-world scenarios where removing stopwords can be harmful

= Although stopwords are often removed, in some cases they are very important.

1) Sentiment Analysis
"I do not like this product"

2) Question Answering / chatbots.
"What is your name?"

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Q.4 Explain the difference between stemming and lemmatization.

= Both techniques are used to reduce words to there base form, but they work differently.

Stemming :

- cuts words using rules.
- May produce non-real words.
- faster but less accurate

Lemmatization :

- Uses vocabulary + grammer rules.
- produces meaningful root words.
- slower but more accurate

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